

Q3.

3.1.  $\rightarrow$

$$W \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$C_1 \Rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = 2C_1$$

$$C_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = 3C_1$$

$\therefore C_1, C_2 \& C_3$  are linearly dependent  $\therefore$  matrix contain only 1 independent direction of info.

3.2  $\rightarrow$

$$\text{Rank } W \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

Echlon form

$$\begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$  Rank  $\Rightarrow$  no. of non zero rows  $\Rightarrow 1$

3.3.

$$\text{Let } C \Rightarrow C_1 \Rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3 \times 1}$$

Columns of  $W$  can be given as L.f.  $1 \cdot C_1, 2 \cdot C_1, 3 \cdot C_1$

$$\therefore R \Rightarrow [1 \ 2 \ 3]_{1 \times 3}$$

$$\therefore \left[ CR \rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [1 \ 2 \ 3] \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} = W \right]$$

Dimension of  $C \Rightarrow 3 \times 1$

Dimension of  $R \Rightarrow 1 \times 3$